

Mobile App & Backend development practical

Overview

This is a filter interview, and only if you demonstrate that you possess good practical solid skills will you be considered for the next phase of the interview.

You are required to complete the assignment and post it in your GitHub repository. **Share the links with us for review by 4pm Friday Oct 24, 2025.** Upon successful review, you will be invited to a virtual session to demonstrate your solution and answer follow up questions.

The assignment is broken down into two parts:

1. Mobile front-end development (60% Score)
2. Backend development (40% Score)

Mobile front end development

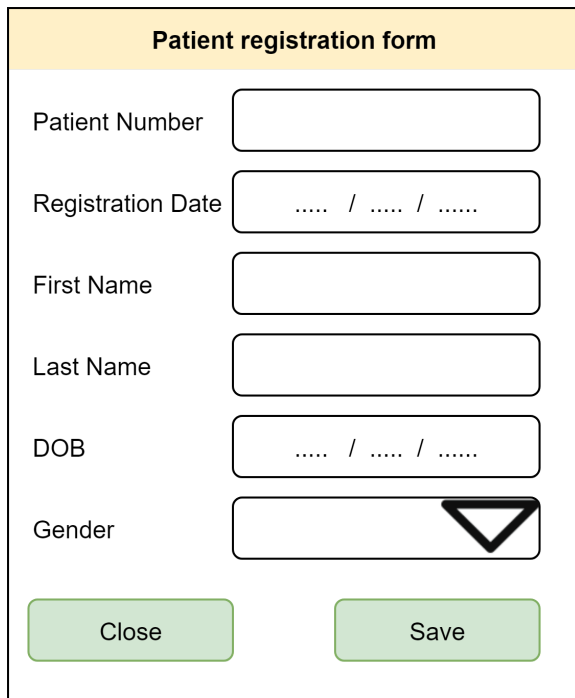
Assignment

Develop an android application to register patients, take vitals (Height & Weight and calculate the BMI), collect additional information via two custom forms and list the patient details. The application will contain five(5) pages, namely:

- Patient registration page
- Vitals form
- Form A
- Form B
- Patient Listing

Patient registration page

- Screen Mockup



The mockup shows a 'Patient registration form' with a yellow header. It contains six input fields: 'Patient Number' (text), 'Registration Date' (date in MM/DD/YYYY format), 'First Name' (text), 'Last Name' (text), 'DOB' (date in MM/DD/YYYY format), and 'Gender' (dropdown menu with a downward arrow). At the bottom are two green buttons: 'Close' and 'Save'.

- It contains a **Registration form** that captures the following fields (3):
 - Patient Id (This is a unique field and cannot be shared between patients)
 - Registration date
 - First name
 - Last name
 - Date of birth
 - Gender

- A patient can only be registered once (1)
- Upon saving the Patient Registration details, load the Vitals page(1)

Vitals page

- Screen Mockup

Patient Vitals form

Patient Name:

Visit Date:

Height:

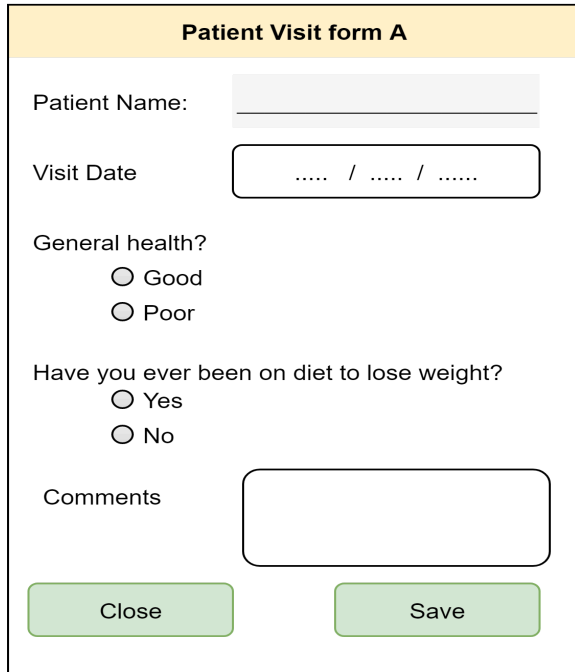
Weight:

BMI:

- It contains a **Patient Vitals form** that captures the following fields:
 - Visit Date
 - Height (In CM)
 - Weight (In KG)
 - BMI (This value is auto-calculated based on Height & Weight: ie. ***BMI = Weight(kg)/ Height(M)²***)
- All fields are mandatory
- A single patient can have multiple submissions for the vitals form but on different dates.
- Upon saving the vitals form, if the patient's BMI (BMI = kg/m²) is less or equal to 25, the app should automatically load Visit Page B; otherwise, load Visit Page A.

Visit Page A

- Mockup



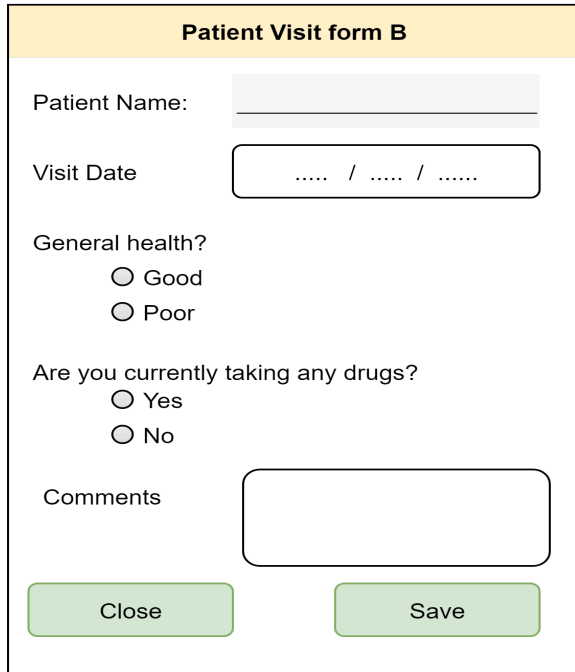
The mockup shows a form titled "Patient Visit form A" with a yellow header. The form contains the following fields and controls:

- Patient Name:** A text input field.
- Visit Date:** A date input field with a placeholder ".... / /"
- General health?:** Two radio button options: "Good" and "Poor".
- Have you ever been on diet to lose weight?:** Two radio button options: "Yes" and "No".
- Comments:** A large text area.
- Buttons:** "Close" and "Save" buttons at the bottom.

- Contains **Visit Form A**, which collects the following information:
 - Visit Date
 - General health (Good/Poor)
 - Have you ever been on a diet to lose weight? (Yes/No)
 - Comments
- A single patient can have multiple submissions for form A but on different dates.
- The page is only visible if BMI < 25
- All fields are mandatory
- Upon saving this form, load the patient listing page

Visit Page B

- Screen Mockup



The mockup shows a form titled "Patient Visit form B" with a yellow header. The form contains the following fields and controls:

- Patient Name:** A text input field.
- Visit Date:** A date input field with a placeholder ".... / /"
- General health?:** Two radio button options: "Good" and "Poor".
- Are you currently taking any drugs?:** Two radio button options: "Yes" and "No".
- Comments:** A large text area for notes.
- Buttons:** "Close" and "Save" buttons at the bottom.

- Contains **Visit Form B** which collects the following information:
 - Visit Date
 - General Health (Good/Poor)
 - Are you currently using any drugs (Yes/No)
 - Comments
- A single patient can have multiple submissions for Visit form B but on different dates.
- Should only be displayed for patients whose BMI is **equal to or greater than 25**
- Upon saving the form, load the patient listing page

Patient Listing Page

- Screen Mockup

Patient Listing		
Date	 / /	
Patient Name	Age	BMI Status
John Doe	35	Normal
Jane Doe	18	Overweight
James Doe	20	Underweight
Julia Doe	28	Normal

- Should display a list of patients showing the following columns:
 - Patient Name
 - Age
 - Last BMI Status
- Let the BMI status be either “Underweight”, “Normal”, or “Overweight”.
Underweight patients have a BMI less than 18.5; Normal patients have a BMI greater than 18.5 and less than 25, and greater or equal to 25 for overweight patients.
- The report should have the ability to filter records by the visit date.

Form submission

Upon saving the forms, the application should:

- Store data locally (Room/SQLite database)
- Trigger data submission to a backend service that you will define in the next task.

Preferred Tech stacks

Java/Kotlin

We will award bonus points for:

- Properly designed, functional and attractive UI

Backend development

Assignment

Design and build a RESTful backend service capable of holding the information collected during the previous phase. The service should support the following end-points:

- 1) Patient Registration
- 2) Vitals
- 3) Visit Forms

Preferred tech stack

Use any of the following Tech stacks:

Database

- MySQL, PostgreSQL, MongoDB

API

- Consider Java Spring Boot or the technology of your choice.

We will award bonus points for a thoughtfully designed backend.